

 Marwadi University Marwadi Chandarana Group	NAAC A+	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Compute and analyze the correlation between three audio tracks	
Experiment: 7	Date:	Enrollment No: 92400133150

Aim :- Image Processing

Theory:-

Image enhancement is an important technique in Digital Image Processing used to improve the quality and visibility of image details. Image negation reverses the intensity values of pixels, where dark pixels become bright and bright pixels become dark. Thresholding converts a grayscale image into a binary image by comparing each pixel with a selected threshold value. In this experiment, a threshold value of 128 is used to separate pixels into black and white regions. Gamma correction is a nonlinear intensity transformation used to adjust the brightness of an image. It uses the gamma value to modify the intensity distribution and improve the visibility of certain details. These techniques are commonly used in image enhancement, segmentation, object detection, and image analysis.

Image enhancement and transformation techniques are used in Digital Image Processing to modify an image and make important features easier to analyze or observe. In this program, three basic image processing techniques are applied to a grayscale image: **image negation, thresholding, and gamma correction.**

1. Image Negation

Image negation is used to create the **negative of an image**. It reverses the intensity values of the pixels.

For an 8-bit grayscale image, the intensity values range from **0 to 255**. The negative is calculated using:

$$\text{Negative} = 255 - \text{Original Pixel}$$

Therefore, black pixels become white, white pixels become black, and gray pixels are converted to their corresponding opposite intensity.

Image negation is useful when details that are difficult to observe in the original image become more visible in the negative image.

2. Thresholding

Thresholding is a technique used to convert a grayscale image into a **binary image** containing mainly black and white pixels.

In this program, the threshold value is set to **128**. Pixels with intensity greater than or equal to 128 are converted to 255 (white), while pixels below 128 are converted to 0 (black).

Thresholding is commonly used for **object detection, image segmentation, document processing, and separating objects from the background.**

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3. Gamma Correction

Gamma correction is an intensity transformation technique used to adjust the brightness of an image.

First, the pixel values are normalized from the range **[0, 255]** to **[0, 1]**. The gamma transformation is then applied using:

$$\text{Output} = \text{Input}^{(1/\gamma)}$$

In this program, the gamma value is:

$$\gamma = 2.0$$

Gamma correction can make dark or bright regions of an image more visible. It is commonly used for **brightness adjustment, image enhancement, display correction, and improving visual details.**

Explanation

Step 1: Import Required Libraries

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

Three libraries are imported.

- **OpenCV (cv2)** is used to read and process the image.
- **NumPy (numpy)** is used for numerical and pixel-value operations.
- **Matplotlib (matplotlib.pyplot)** is used to display the images.

Step 2: Read the Image

```
src_image = cv2.imread("images.jpg", cv2.IMREAD_GRAYSCALE)
```

The image named images.jpg is loaded using OpenCV.

The cv2.IMREAD_GRAYSCALE option directly loads the image as a **grayscale image**, so each pixel contains an intensity value between 0 and 255.

Step 3: Check Whether the Image is Loaded

if src_image is None:

```
    print("Image not found!")
```

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exit()

This checks whether OpenCV successfully loaded the image.

If the image cannot be found or opened, src_image will be None. In that case, the program prints an error message and stops execution.

Step 4: Perform Image Negation

```
negative_image = 255 - src_image
```

This line creates the negative version of the original grayscale image.

For example:

Original pixel = 0

Negative pixel = 255

and:

Original pixel = 255

Negative pixel = 0

Thus, dark areas become bright and bright areas become dark.

Step 5: Perform Thresholding

```
_ , thresholded_image = cv2.threshold(
    src_image, 128, 255, cv2.THRESH_BINARY
)
```

The cv2.threshold() function converts the grayscale image into a binary image.

The values used are:

- **128** → threshold value
- **255** → maximum value
- **cv2.THRESH_BINARY** → binary thresholding method

The basic operation is:

Pixel $\geq$ 128 → 255 (White)

Pixel $<$ 128 → 0 (Black)

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The resulting binary image is stored in thresholded_image.

Step 6: Set the Gamma Value

$\gamma = 2.0$

A gamma value of **2.0** is selected for gamma correction.

The gamma value controls how the brightness of the image is transformed.

Step 7: Normalize the Image

$\text{normalized_image} = \text{src_image} / 255.0$

The original pixel values are in the range:

0 to 255

Dividing by 255.0 converts them to:

0 to 1

This normalized range is required for applying the gamma transformation.

Step 8: Apply Gamma Correction

```
gamma_corrected_image = np.power(
    normalized_image, 1/gamma
)
```

The gamma correction formula is applied using NumPy's np.power() function.

For $\gamma = 2.0$, the exponent becomes:

$1 / 2 = 0.5$

The pixel intensities are therefore transformed according to the gamma correction formula.

Step 9: Convert Back to 0–255 Range

```
gamma_corrected_image = np.uint8(
    gamma_corrected_image * 255
)
```

After gamma correction, the pixel values are still in the range **0 to 1**.

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Multiplying by 255 converts them back to the normal grayscale range:

0 to 255

`np.uint8()` converts the result into an 8-bit unsigned integer image format.

Step 10: Create the Display Window

```
plt.figure(figsize=(15,5))
```

A figure is created with a width of 15 and height of 5.

Four images will be displayed side by side.

Step 11: Display the Original Image

```
plt.subplot(1,4,1)
```

```
plt.imshow(src_image, cmap='gray')
```

```
plt.title("Original")
```

```
plt.axis('off')
```

The original grayscale image is displayed in the **first position**.

`cmap='gray'` ensures that the image is displayed using grayscale colors.

`plt.axis('off')` removes the axis values from the image.

Step 12: Display the Negative Image

```
plt.subplot(1,4,2)
```

```
plt.imshow(negative_image, cmap='gray')
```

```
plt.title("Negative")
```

```
plt.axis('off')
```

The negative image is displayed in the **second position**.

It allows us to visually compare the original image with its intensity-inverted version.

Step 13: Display the Thresholded Image

```
plt.subplot(1,4,3)
```

```
plt.imshow(thresholded_image, cmap='gray')
```

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```
plt.title("Threshold")
```

```
plt.axis('off')
```

The thresholded binary image is displayed in the **third position**.

The image contains mainly black and white regions based on the threshold value of 128.

Step 14: Display the Gamma-Corrected Image

```
plt.subplot(1,4,4)
```

```
plt.imshow(gamma_corrected_image, cmap='gray')
```

```
plt.title("Gamma")
```

```
plt.axis('off')
```

The gamma-corrected image is displayed in the **fourth position**.

This allows us to observe how gamma correction changes the brightness and intensity distribution of the original image.

Step 15: Display All Results

```
plt.show()
```

Finally, plt.show() displays all four images together.

Overall Working

The program first reads a grayscale image and then applies three different image processing techniques. **Image negation** reverses the pixel intensity values, **thresholding** converts the grayscale image into a binary image, and **gamma correction** changes the brightness distribution of the image. Finally, the original image and all three processed images are displayed side by side for easy comparison.

Code:-

```
import cv2
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
# Read image directly in grayscale
```

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```
src_image = cv2.imread('/content/1785219786630.jpg', cv2.IMREAD_GRAYSCALE)
```

```
# Check if image is loaded
```

```
if src_image is None:
```

```
    print("Image not found!")
```

```
    exit()
```

```
# -----
```

```
# 1. Image Negation
```

```
# -----
```

```
negative_image = 255 - src_image
```

```
# -----
```

```
# 2. Thresholding
```

```
# -----
```

```
_, thresholded_image = cv2.threshold(src_image, 128, 255, cv2.THRESH_BINARY)
```

```
# -----
```

```
# 3. Gamma Correction
```

```
# -----
```

```
gamma = 2.0
```

```
# Normalize image to range [0,1]
```

```
normalized_image = src_image / 255.0
```

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Apply gamma correction

```
gamma_corrected_image = np.power(normalized_image, 1/gamma)
```

Convert back to range [0,255]

```
gamma_corrected_image = np.uint8(gamma_corrected_image * 255)
```

Display Images

```
plt.figure(figsize=(15,5))
```

```
plt.subplot(1,4,1)
```

```
plt.imshow(src_image, cmap='gray')
```

```
plt.title("Original")
```

```
plt.axis('off')
```

```
plt.subplot(1,4,2)
```

```
plt.imshow(negative_image, cmap='gray')
```

```
plt.title("Negative")
```

```
plt.axis('off')
```

```
plt.subplot(1,4,3)
```

```
plt.imshow(thresholded_image, cmap='gray')
```

```
plt.title("Threshold")
```

```
plt.axis('off')
```

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```
plt.subplot(1,4,4)
plt.imshow(gamma_corrected_image, cmap='gray')
plt.title("Gamma")
plt.axis('off')
plt.show()
```

Observation

Observation

In this experiment, the original grayscale image was processed using image negation, thresholding, and gamma correction. The negative image showed reversed intensity values, where the dark areas became bright and the bright areas became dark. In the thresholded image, the grayscale image was converted into black and white using a threshold value of 128, making the separation between different regions more noticeable. The gamma-corrected image showed a change in brightness and intensity distribution because a gamma value of 2.0 was applied. By comparing all the output images with the original image, it was clear that each technique changes the image in a different way. Negation is useful for highlighting certain details, thresholding is useful for separating objects from the background, and gamma correction is useful for adjusting image brightness. Overall, the experiment helped in understanding how changing pixel intensity can affect the appearance and analysis of a digital image.

Expected Output:-

The program will display four grayscale images side by side: the Original image, the Negative image, the Thresholded image, and the Gamma-corrected image. The negative image will show reversed intensity values, the thresholded image will contain mainly black and white regions based on the threshold value 128, and the gamma-corrected image will show a change in brightness and intensity using $\gamma = 2.0$. This allows easy visual comparison of the effects of each image processing technique.

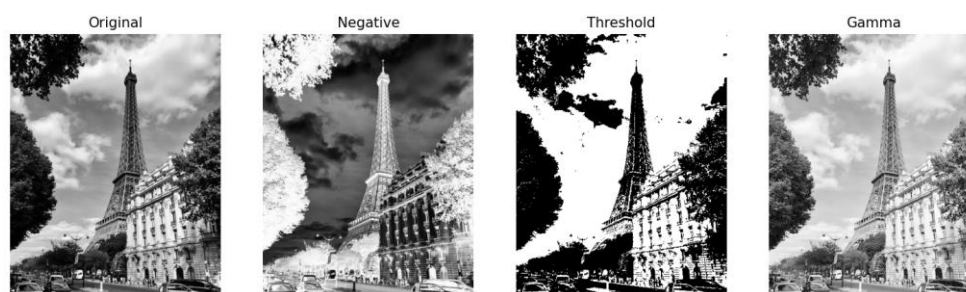
The program displays four images side by side:

1. Original – the input grayscale image.
2. Negative – the intensity-inverted version of the image.
3. Threshold – a black-and-white binary version using a threshold of 128.
4. Gamma – the image after gamma correction with $\gamma = 2.0$.

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The output makes it easy to visually compare how each image processing technique changes the original image.

Output:-



Conclusion :-

From this experiment, I learned how different image intensity transformation techniques can be applied to a grayscale image to change its appearance and improve the visibility of important features. Image negation reversed the intensity values of the original image, causing dark regions to appear bright and bright regions to appear dark. This can be useful when certain details are easier to identify in a negative image. Thresholding converted the grayscale image into a binary image using a threshold value of 128, which helped separate lighter and darker regions and can be useful for image segmentation and object detection. Gamma correction with $\gamma = 2.0$ changed the brightness and intensity distribution of the image using a nonlinear transformation. By comparing the original image with the negative, thresholded, and gamma-corrected images, it was clear that each technique has a different purpose and produces a different visual effect. Overall, this experiment provided a practical understanding of how pixel intensity values can be manipulated to enhance, segment, and analyze digital images, and showed that selecting the appropriate image processing technique depends on the type of image and the desired result.

Github link :-

<https://github.com/manav-chotai081/DSIP/tree/main/Experiment%207>